Flyhomes BBYS Calculator

How It Works — A Plain-Language Logic Guide

Written for product, operations, and business stakeholders — no code required · Reflects the calculator as of 2026-08-13

What this document is. A walkthrough of the decision logic behind the BBYS ("Buy Before You Sell") calculator: which product gets recommended for which combination of buyer challenges, and exactly how each number on the results screen — Estimated Max Loan Amount, Estimated Cost, and Estimated Benefit from BBYS — gets calculated. Every example in this document uses the calculator's own default inputs so the math can be checked against the live tool.

Baseline example used throughout this document (the calculator's own defaults):
Current home value: **$1,200,000** · Current mortgage balance: **$300,000** · Purchase price: **$800,000** · Cash & liquid assets: **$40,000** · Transition days: **60**

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1. The four products, in plain terms

Every recommendation the calculator makes is built from four underlying Flyhomes products. Everything else in this document is really about "which of these gets picked, and how much does it cost."

DTI BUSTER **Guaranteed Backup Contract (GBC)**

Flyhomes guarantees to buy the client's current home if it doesn't sell in time. This lets the client set their current mortgage aside when qualifying for a new loan (it's no longer counted against their debt-to-income ratio) and drop the home-sale contingency on their offer. No new loan is issued — it's a contract, not financing, so it's the cheapest of the four options.

INSTANT EQUITY **Equity for Down Payment**

Unlocks equity from the client's current (not-yet-sold) home so it can be used toward the down payment and closing costs on the new home, before the old one sells.

CASH OFFER **All-Cash Advantage**

Flyhomes fronts the money so the client can make a cash-strength offer — no financing or sale contingency, closing in as little as 10 days — competing like an all-cash buyer against other bidders.

CROSS COLLATERAL **Move with $0 Out of Pocket / Retire & Downsize**

Combines the equity in *both* the current and new home into one loan, so the client can move with little or no cash out of pocket. Often eliminates the need for a new long-term mortgage entirely — this is also the product behind the "Retire & Downsize" scenario. Requires both homes to be in the same state.

How a buyer tells the calculator what they need

Every recommendation starts from up to four boxes a buyer can check, describing what's getting in their way:

#	Challenge option	What it signals
1	"My current mortgage is capping how much I qualify for"	DTI (debt-to-income) problem
2	"My cash is tied up in my current home"	Liquidity / down-payment problem
3	"My offers keep losing to cash buyers"	Competitiveness / speed problem
4	"I want to downsize — ideally without a new mortgage"	Retirement / downsizing intent

A buyer can check any combination — including none, or all four. §2 covers what happens for every combination.

2. What gets recommended for each combination of challenges

Checking Option 4 ("want to downsize") always wins outright, no matter what else is checked — so it gets its own row. Every other combination of Options 1–3 is listed below.

Opt 1 DTI	Opt 2 Equity	Opt 3 Cash offer	Opt 4 Downsize	Recommended solution
—	—	—	✔	CROSS COLLATERAL "Retire & Downsize" — regardless of Options 1–3[1]
—	—	—	—	Nothing recommended (nothing selected)
✔	—	—	—	DTI BUSTER alone — no loan product, no other cards shown
—	✔	—	—	INSTANT EQUITY + DTI BUSTER *suggested*[2]
—	—	✔	—	CROSS COLLATERAL *or* CASH OFFER + DTI BUSTER *suggested* — depends on the buyer's numbers[3]
✔	✔	—	—	INSTANT EQUITY + DTI BUSTER (fully priced — Option 1 is checked)
✔	—	✔	—	CROSS COLLATERAL *or* CASH OFFER + DTI BUSTER (fully priced) — depends on the buyer's numbers[3]
—	✔	✔	—	CROSS COLLATERAL — always, regardless of the numbers[4]
✔	✔	✔	—	CROSS COLLATERAL — Option 1 doesn't change this[4]

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1. Unless the current home and the new home are in different states — see §3, "Cross Collateral requires same state."

2. "Suggested" is an important distinction: the DTI Buster card is shown with a "Most likely needed" badge, but its fee is not added to Estimated Cost unless the buyer also manually checks Option 1. Checking Option 1 makes it real (priced in) and removes the badge.

3. The calculator projects the buyer's final loan-to-value on the new home. If that projection comes out at 75% or below, Cross Collateral is the better fit and is recommended alone. Above 75%, Cash Offer + DTI Buster is recommended instead (DTI Buster still follows the "suggested vs. priced" rule from note 2). A buyer can also manually request Cash Offer over Cross Collateral via a "Keep More Cash" toggle on the results screen.

4. Cross Collateral can never be paired with DTI Buster (GBC) — by design, the two products don't combine, so checking Option 1 alongside Options 2+3 has no effect on the recommendation or the price.

3. Rules that can override the default recommendation

Cross Collateral requires both homes in the same state

If the buyer's current-home state and new-home state are different, Cross Collateral is never recommended — no matter which combination of options would otherwise have triggered it. Instead:

What would have been recommended	What's recommended instead when states differ
Retire & Downsize (Option 4)	INSTANT EQUITY + DTI BUSTER (fully priced)
Cross Collateral from Options 2+3	CASH OFFER + DTI BUSTER (fully priced)

Why the difference: downsizing alone doesn't signal "I need a competitive cash offer" — it signals "I want to use my equity" — so it falls back to the product that best matches that intent. Options 2+3 together explicitly ask for a competitive cash offer, so that's what it falls back to.

Home value above $2,000,000 isn't estimated

Current home values above $2,000,000 fall outside the standard DTI Buster fee tiers and require manual pricing. Rather than show an incomplete number, the calculator doesn't attempt an estimate at all above this value — the results panel is hidden and the buyer sees a "please contact us" message instead. The input slider is capped at $2,000,000 for this reason.

"Keep More Cash" manual override

When the numbers point to Cross Collateral (Options 2+3, or the Option 3 low-LTV case), the results screen shows a banner letting the buyer manually switch to Cash Offer instead — for buyers who'd rather carry a bit more of a mortgage in exchange for keeping more cash from the sale. Switching this way does not add DTI Buster to the quote, even if Option 1 was checked, since Cross Collateral (what the buyer is switching away from) never carried a DTI Buster price to begin with.

Low-cash warning

When Cash Offer is the resolved recommendation but the buyer's combined cash + unlockable equity can't cover 5% of the purchase price, a warning banner shows the exact shortfall (e.g. "You need $12,000 more in cash or home equity") rather than silently recommending a plan the buyer can't actually afford.

4. How Estimated Max Loan Amount is calculated

Two numbers feed almost every formula in this section:

Available Equity = Current Home Value × 67.5%* − Current Mortgage Balance

Combined Resources = Cash & Liquid Assets + Available Equity

* 67.5% = 75% loan-to-value on the home × a 90% safety cushion.

Baseline example

Available Equity = $1,200,000 × 67.5% − $300,000 = **$510,000**

Combined Resources = $40,000 + $510,000 = **$550,000**

INSTANT EQUITY

Max Loan Amount = Available Equity

Example

Max Loan Amount = **$510,000**

CASH OFFER

Max Purchase Price the buyer can qualify for = Combined Resources ÷ 5%

Max Loan Amount = min(Purchase Price, Max Purchase Price) × 95%

Example (with $1,000,000 current mortgage, to illustrate this branch)

Available Equity is $0 here (mortgage exceeds the equity formula) → Combined Resources = $40,000

Max Purchase Price = $40,000 ÷ 5% = $800,000

Max Loan Amount = min($800,000, $800,000) × 95% = **$760,000**

CROSS COLLATERAL

Combined-equity cap = (Purchase Price + Current Home Value) × 75% − Current Mortgage Balance

Max Loan Amount = min(Purchase Price × 105%, Combined-equity cap)

Baseline example

Combined-equity cap = ($800,000 + $1,200,000) × 75% − $300,000 = $1,200,000

Max Loan Amount = min($800,000 × 105% = $840,000, $1,200,000) = **$840,000**

DTI BUSTER alone

No loan is issued — DTI Buster is a backup-purchase guarantee, not financing. There is no Max Loan Amount for this scenario, and the "Estimated Max Loan Amount" card isn't shown at all.

5. How Estimated Cost is calculated

Estimated Cost has up to three parts, all rolled into one number on the results screen (with a breakdown available on tap):

```
Estimated Cost = One-Time Fees + Accrued Interest
```

"One-Time Fees" itself bundles three things the buyer never sees broken out individually — an origination fee, a broker fee, and (when it applies) the DTI Buster fee:

Origination fee

Product	Rate
INSTANT EQUITY	2.0% of the loan amount, always
CASH OFFER / CROSS COLLATERAL	1.5% if the loan is above 90% of the purchase price, otherwise 1.0%

Broker fee

Product	Rate
INSTANT EQUITY	0.5% of the loan amount
CASH OFFER	1% of the loan amount
CROSS COLLATERAL	1% of the loan amount

DTI BUSTER fee

Priced by tier based on the current home's value. When it's paired with a Flyhomes loan product (any scenario except a pure DTI-only recommendation), a bundle credit applies — the buyer pays less than they would standalone.

Current home value	Standalone fee (DTI Buster alone)	Bundled fee (paired with a loan)
Up to $500,000	$2,500	$2,500
$500,001 – $750,000	$3,500	$2,500
$750,001 – $1,000,000	$5,000	$2,500
$1,000,001 – $1,500,000	$7,500	$5,000
$1,500,001 – $2,000,000	$10,000	$7,500

Above $2,000,000: not estimated at all — see §3.

Accrued interest

```
Accrued Interest = Loan Amount × 9.99% × (Transition Days ÷ 365)
```

Only applies when there's an actual loan (not for DTI Buster alone). No monthly payments — it's settled in one lump sum when the current home sells. "Transition Days" defaults to 60 and is editable by the buyer.

Worked examples (baseline scenario, 60 transition days)

Scenario	Loan amount	Origination	Broker	DTI Buster	Accrued interest	Estimated Cost
INSTANT EQUITY alone	$510,000	$10,200	$2,550	—	$8,375	**$21,125**
INSTANT EQUITY + DTI BUSTER	$510,000	$10,200	$2,550	$5,000	$8,375	**$26,125**
DTI BUSTER alone	—	—	—	$7,500[†]	—	**$7,500**
CROSS COLLATERAL	$840,000	$12,600	$8,400	—	$13,794	**$34,794**
CASH OFFER (higher-LTV scenario)	$760,000	$11,400	$7,600	—	$12,481	**$31,481**
CASH OFFER + DTI BUSTER	$760,000	$11,400	$7,600	$5,000	$12,481	**$36,481**

[†] Standalone rate, since DTI Buster alone has no loan to bundle against.

6. How Estimated Benefit from BBYS is calculated

This is the buyer-facing "here's what going through Flyhomes saves you" number, made up of up to four pieces plus a "move once, not twice" logistics estimate. Every slider below is editable by the buyer; defaults shown.

1. Cash Offer Price Advantage (default 3.5%, editable 1–10%)

```
Purchase Price × Cash Discount %
```

A cash-strength offer can often win at a lower price than a financed one has to bid. Shown only when the recommendation involves a cash offer or Cross Collateral / Retire & Downsize.

Example

$800,000 × 3.5% = **$28,000**

2. Staged Home Premium (default 5%, editable 1–10%)

```
max(0, Current Home Value × Staged Premium % − $3,500 staging cost)
```

Staging an empty home can lift its sale price; this nets out a typical $3,500 staging cost.

Example

$1,200,000 × 5% − $3,500 = **$56,500**

3. Home Price Appreciation (default 4%/yr, editable −10% to 10%)

```
Purchase Price × (HPA Rate ÷ 12) × (Transition Days ÷ 30)
```

Buy-side only — this is the price increase avoided by buying now instead of waiting until after the sale. (Flyhomes always recommends selling the current home as soon as possible regardless.)

Example (60 transition days)

$800,000 × (4% ÷ 12) × (60 ÷ 30) = **$5,333**

4. "Move Once, Not Twice" — logistics

Three auto-computed (and editable) line items, based on the current home's value and transition days:

```
Extra Moving Fee = Current Home Value × 0.3% + $500
Temporary Housing = Current Home Value × 0.6% × (Transition Days ÷ 30)
Storage = ($150 + Current Home Value × 0.02%) × (Transition Days ÷ 30)
```

Example (60 transition days)

Extra Moving Fee = $1,200,000 × 0.3% + $500 = $4,100
Temporary Housing = $1,200,000 × 0.6% × 2 = $14,400
Storage = ($150 + $240) × 2 = $780
Logistics total = **$19,280**

Grand total

Estimated Benefit = Cash Offer Advantage + Staged Home Premium + HPA + Logistics

Baseline example, cash offer active

$28,000 + $56,500 + $5,333 + $19,280 = **$109,113**

Baseline example, DTI Buster alone (no cash offer row)

$56,500 + $5,333 + $19,280 = **$81,113**

7. Quick reference: all rates and thresholds

Item	Value
Instant Equity multiplier	67.5% of current home value (75% LTV × 90% cushion), minus current mortgage
Cash Offer max loan-to-value	95% of purchase price
Cash Offer minimum down payment	5% of purchase price (from cash + unlockable equity combined)
Cross Collateral acquiring LTV cap	105% of purchase price
Cross Collateral combined LTV cap	75% of (purchase price + current home value)
Origination fee — Cash Offer / Cross Collateral	1.5% if loan > 90% of purchase price, else 1.0%
Origination fee — Instant Equity	2.0%, always
Broker fee — Instant Equity	0.5% of loan amount
Broker fee — Cash Offer / Cross Collateral	1.0% of loan amount
Accrued interest rate	9.99% per year, prorated by transition days
Rule 3 cash-vs-CC threshold	Projected final LTV ≤ 75% → Cross Collateral; above → Cash Offer
Cross-state Cross Collateral block	Blocked whenever current-home state ≠ new-home state
Home value input ceiling	$2,000,000 — no estimate above this value
Staging cost (flat)	$3,500, editable
Default Estimated Benefit sliders	Cash discount 3.5%, Staged premium 5%, HPA 4%/yr
Default transition days	60 (minimum 1), editable by buyer

A plain-language companion to the engineering handoff spec (`BBYS-Calculator-Logic-Spec.pdf`). Every example number in this document was checked against the live calculator on 2026-08-13. Rates and thresholds will drift as the product evolves — re-verify before relying on this for pricing decisions.